BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section-A Multiple Choice Questions (MCQ's)

Note: Q.1 Choose the correct answer for each from the given options.

Q1. For two sets A and B, if O(A) = O(B) then

(a) A B
(b) B = A
(c) A B
(d) A c B

Q2. ((A')') = _____

(a) (A')'
(b) A'
(c) A
(d) B

Q3. Which operation is not commutative

(a) Symmetric difference
(b) Union
(c) Difference
(d) Intersection

Q4. Which one is subset of N

(a) Z
(b) Q
(c) E
(d) P

Q5. Tabular form of A = {x | x ∈ Z and x = 25} is

(a) {1,5}
(b) {-5,5}
(c) {1,5,25}
(d) {-5}

Q6. One and only one circle passes through three points.

(a) collinear
(b) non-collinear
(c) disjoint
(d) none of these

Q7. The hypothesis of the enunciation "If two chords of a circle are congruent, then they are equidistant from the centre." is:

(a) two chords of a circle are equidistant from the centre.
(b) two chords of a circle are congruent.
(c) a circle has two chords.
(d) the centre of circle is equidistant from chords.

Q8. The hypothesis of the enunciation "One and only one circle can pass through three non-collinear points" is:

(a) three points are non-collinear.
(b) one and only one circle passes through three points.
(c) two circles pass through three points.
(d) three points are collinear.

Q9. The conclusion of the enunciation "One and only one circle can pass through three non-collinear points" is:

(a) three points are non-collinear.
(b) one and only one circle passes through three points.
(c) two circles pass through three points.
(d) three points are collinear.

Q10. A circle is an example of a curve

(a) simple and closed
(b) simple and open
(c) non-simple and closed
(d) non-simple and open

Q11. The conclusion of the enunciation "If two chords of a circle are congruent, then they are equidistant from the centre." is:

(a) two chords of the circle are equidistant from the centre.
(b) two chords of a circle are congruent.
(c) a circle has two chords.
(d) the centre of circle is equidistant from chords.

Q12. In the adjacent figure, the point of tangency is .

(a) A
(b) B
(c) C
(d) D

Q13. Tangents drawn at end points of a diameter of a circle are __ to each other

(a) parallel
(b) perpendicular
(c) intersecting
(d) both (b) and (c)

Q14. The maximum number of common tangents between two circles touching internally __ is

(a) 0
(b) 1
(c) 2
(d) 3

Q15. The maximum number of common tangents between two circles touching externally is

(a) 0 (b) 1

(c) 2 (d) 3

Q16. If a point is outside the circle then from this point we can draw tangent(s) to the circle.

(a) one (b) two

(c) three (d) none

Q17. Angle between the radial segment and tangent at its outer end point is

(a) 45 degrees (b) 60 degrees

(c) 90 degrees (d) 120 degrees

Q18. In the adjacent figure, circles with centres E and C

(a) touch internally (b) touch externally

(c) do not touch (d) are congruent

Q19. In the adjacent figure, circles with centres B and C have _____ point(s) of contact.

(a) no (b) one

(c) two (d) none of these

Q20. In adjacent figure, circles with centres E and C have point(s) of contact.

(a) no (b) one

(c) two (d) none of these

Q21. In the adjacent figure, AB is

(a) tangent (b) secant

(c) chord (d) none of these

Section-B

> Note: Attempt any 08 question of the following:

Q2. Write any five example of empty sets.

Q3. Which of the following sets are disjoint, overlapping, exhustive and cells. (i) {1,2,3,5,7} and {4,6,8,9,10} (ii) {1,2,3,6} and {1,2,4,8}

Q4. If, A and B are subset of U then prove the following using properties. (i) AU(A∩B)=A∩(AUB) (ii) AUB=AU(A'∩B) (iii) B=(A∩B)U(A'∩B) (iv) B=AU(A'∩B), if A subset B

Q5. If a : b = 5 : 8, find the vlaue of 3a + 4b: 5a + 7b

Q6. Find the value of x in the following contained prppotions. (i) 45,x , 5 (ii) 7, x - 3 , 112

Q7. Solve the equations: (i) ((x+3)^2 - (x-5)^2)/((x+3)^2 + (x-5)^2) = 4/5 (ii) (sqrt(x+1)+sqrt(x-1))/(sqrt(x+1)-sqrt(x-1)) = 1/2

Q8. If y varies directly as x2 and z and y = 6 when x = 4, z = 9. Write y as a function of x and z determine the value of y when x = -8, and z = 12.

Q9. If p:q = r:s then show that (p^2 + q^2): p^3/(p+q) = (r^2 + s^2): r^3/(r+s)

Q10. For the matrices A = [[-1, 0, 1],[2, 1, 0],[3, 2, -1]] and B = [[4, 2, 3],[-2, 4, 1],[3, 2, 1]]. Find (i) A + B (ii) A - B (iii) 3A + 2B (iv) AB (v) BA (vi) A^2

Q11. For what value of x, the matrix [[5 - x, x + 1],[2, 4]] is singular?

Q12. Construct histogram for the following date with unegull class intervals: Class limits: 0-29, 40-49, 50-79, 80-99; Frequencies: 9, 8, 12, 4

Section-C

> Note: Attempt any 04 Question of the following:

Q13. Resolve the follwoing fractions into partial functions: (I) (9x^2 + 5x + 7)/(x(x+2)(x-5)) (ii) (x+5)/((x^2+1)^2 (x-3))

Q14. In triangle ABC, PQ parallel BC. Find x if mAP = 5x - 3, mBP = 2, mPB = 2, mAQ = 2x +1 and mQC = 3.

Q15. Ratio of corresponding sides of two similar triangle is 2: x : x - 5 and the ratio of their areas is 1:9. Find the vlaue of x.

Q16. Show that one and only once circle can pass through the vertices of a square.

Q17. Show that the diameters of a circle bisect each other.